Project 5 — Full Network Assessment

Network Scanning and Enumeration — Final Project Report

Student Name	Steve
Project	Project 5 — Full Network Assessment
Specialization	Network Scanning and Enumeration
Scanner Machine	Kali Linux (hostname: V2310)
Target 1	192.168.10.71 — Windows Server VM (V2434 / STEVE), Active Directory domain controller for steve.local
Target 2	192.168.10.64 — Windows 10 VM (Seemi-s-A07) — note: same VM as the earlier 192.168.10.33; its DHCP address changed
Target 3	192.168.10.91 — Metasploitable 3, Ubuntu 14.04 LTS VM
Tools Used	Nmap (host discovery, port scan, service/version + OS detection, NSE vuln scripts)

1. Objective

The objective of this project is to perform a full network security assessment of three lab machines from Kali Linux: discover which hosts are online, map every open port and running service with versions and operating systems, check the services for known vulnerabilities with Nmap's vuln scripts, map the findings, and finish with a remediation roadmap — the list of fixes each finding needs.

2. Tools Used

Tool	Used For
Nmap 7.991 (Kali Linux)	Ping checks, port scans, service/version detection (-sV), OS detection (-O), and vulnerability checking with the NSE vuln script category
Linux grep	Searching the long vulnerability-scan output for VULNERABLE findings instead of reading thousands of lines by hand

3. Procedure

Step 1 — Host discovery (on Kali).

All three target machines were confirmed online with ping before scanning: 192.168.10.71, 192.168.10.64 and 192.168.10.91 all answered. (The Windows 10 VM had a new DHCP address: 192.168.10.64 instead of the earlier 192.168.10.33; its MAC address confirmed it is the same machine.)

```
ping -c 2 192.168.10.71
```

```
ping -c 2 192.168.10.64
```

```
ping -c 2 192.168.10.91
```

Step 2 — Port scan of all three targets (on Kali).

One scan mapped the open ports of every target at once. The results became the target list for the deeper scans.

```
nmap 192.168.10.71 192.168.10.64 192.168.10.91 -oN p5-ports.txt
```

```
Session  Actions  Edit  View  Help
┌──(root㉿V2310)-[/home/steve]
└─# nmap 192.168.10.71 192.168.10.64 192.168.10.91 -oN p5-ports.txt
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-21 06:42 -0400
Nmap scan report for V2434 (192.168.10.71)
Host is up (0.00039s latency).
Not shown: 984 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
111/tcp  open  rpcbind
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
2049/tcp open  nfs
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5985/tcp open  wsman
MAC Address: 00:0C:29:C4:FA:41 (VMware)

Nmap scan report for Seemi-s-A07 (192.168.10.64)
Host is up (0.00071s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 00:0C:29:B0:A0:2B (VMware)

Nmap scan report for metasploitable3-ub1404 (192.168.10.91)
Host is up (0.00089s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT     STATE  SERVICE
21/tcp   open   ftp
22/tcp   open   ssh
80/tcp   open   http
445/tcp  open   microsoft-ds
631/tcp  open   ipp
3000/tcp closed ppp
3306/tcp open   mysql
8181/tcp closed intermapper
MAC Address: 00:0C:29:08:C5:97 (VMware)

Nmap done: 3 IP addresses (3 hosts up) scanned in 5.84 seconds

┌──(root㉿V2310)-[/home/steve]
└─#
```

Figure 1 — Step 2: Port scan of all three targets

What this screenshot shows: All three hosts are up. 192.168.10.71 shows the same 16 open ports as in the earlier projects (21, 53, 80, 88, 111, 135, 139, 389, 445, 464, 593, 636, 2049, 3268, 3269, 5985), with 984 ports closed. 192.168.10.64 (the Windows 10 VM, hostname Seemi-s-A07) has 3 open ports — 135, 139, 445 — and 997 closed. 192.168.10.91 has 6 open ports — 21, 22, 80, 445, 631, 3306 — with 3000 and 8181 closed and 992 filtered; port 8080, open in an earlier exercise, is filtered now because its firewall allow rule was accidentally removed during the Project 4 cleanup. The whole scan took 5.84 seconds.

Step 3 — Service, version and OS detection (on Kali).

Each open port was fingerprinted so the assessment records not just the port, but the exact service and version running on it, plus the operating system.

```
sudo nmap -sV -O 192.168.10.71 192.168.10.64 192.168.10.91 -oN p5-services.txt
```

```
└─# sudo nmap -sV -O 192.168.10.71 192.168.10.64 192.168.10.91 -oN p5-services.txt
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-21 06:44 -0400
Nmap scan report for V2434 (192.168.10.71)
Host is up (0.00069s latency).
Not shown: 984 closed tcp ports (reset)
PORT     STATE SERVICE      VERSION
21/tcp   open  tcpwrapped
53/tcp   open  domain       Simple DNS Plus
80/tcp   open  http         Microsoft IIS httpd 10.0
88/tcp   open  kerberos-sec Microsoft Windows Kerberos (server time: 2026-10-21 10:44:29Z)
111/tcp  open  rpcbind      2-4 (RPC #100000)
135/tcp  open  msrpc?
139/tcp  open  netbios-ssn?
389/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: steve.local, Site: Def
ault-First-Site-Name)
445/tcp  open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds (workgroup: STEVE0
)
464/tcp  open  kpasswd5?
593/tcp  open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp  open  tcpwrapped
2049/tcp open  nlockmgr     1-4 (RPC #100021)
3268/tcp open  ldap         Microsoft Windows Active Directory LDAP (Domain: steve.local, Site: Def
ault-First-Site-Name)
3269/tcp open  tcpwrapped
5985/tcp open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
1 service unrecognized despite returning data. If you know the service/version, please submit the f
ollowing fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port139-TCP:V=7.991%I=7%D=10/21%Time=6AD8978F%P=x86_64-pc-linux-gnu%r(G
SF:etRequest,5,"\x83\0\0\x01\x8f");
MAC Address: 00:0C:29:C4:FA:41 (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.991%E=4%D=10/21%OT=53%CT=1%CU=40181%PV=Y%DS=1%DC=D%G=Y%M=000C29
OS:%TM=6AD897A3%P=x86_64-pc-linux-gnu)SEQ(SP=103%GCD=1%ISR=10E%TI=I%CI=I%TS
OS:=A)SEQ(SP=104%GCD=1%ISR=10B%TI=I%CI=I%TS=A)SEQ(SP=106%GCD=1%ISR=10D%TI=I
OS:%CI=I%II=I%SS=S%TS=A)SEQ(SP=108%GCD=1%ISR=109%TI=I%CI=I%TS=A)SEQ(SP=FE%G
OS:CD=1%ISR=109%TI=I%CI=I%TS=A)OPS(O1=M5B4NW8ST11%O2=M5B4NW8ST11%O3=M5B4NW8
OS:NNT11%O4=M5B4NW8ST11%O5=M5B4NW8ST11%O6=M5B4ST11)WIN(W1=2000%W2=2000%W3=2
OS:000%W4=2000%W5=2000%W6=2000)ECN(R=Y%DF=Y%T=80%W=2000%O=M5B4NW8NNS%CC=Y%Q
OS:=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=
OS:AR%O=%RD=B1C2A1A3%Q=)T3(R=Y%DF=Y%T=80%W=0%S=Z%A=O%F=AR%O=%RD=B1C2A1A3%Q=
OS:)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=B1C2A1A3%Q=)T5(R=Y%DF=Y%T=80%W=0
OS:%S=Z%A=S+%F=AR%O=%RD=B1C2A1A3%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=
OS:B1C2A1A3%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=B1C2A1A3%Q=)U1(R=Y%
OS:DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=N)IE(R=Y%D
OS:FI=N%T=80%CD=Z)

Network Distance: 1 hop
Service Info: Host: STEVE; OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for Seemi-s-A07 (192.168.10.64)
Host is up (0.00077s latency).
```

Figure 2 — Step 3: Service details of 192.168.10.71

What this screenshot shows: 192.168.10.71 is a Windows Server acting as an Active Directory domain controller: DNS (Simple DNS Plus), IIS 10.0 on HTTP, Windows Kerberos, Active Directory LDAP on 389 and 3268 explicitly naming Domain: steve.local, microsoft-ds identified as Windows Server 2008 R2 - 2012, and HTTPAPI 2.0 on 5985. Service Info reports Host: STEVE, OS: Windows (no exact OS match, which is normal for Windows targets).

```
OS:%S=Z%A=S+%F=AR%O=%RD=B1C2A1A3%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=
OS:B1C2A1A3%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=B1C2A1A3%Q=)U1(R=Y%
OS:DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=N)IE(R=Y%D
OS:FI=N%T=80%CD=Z)

Network Distance: 1 hop
Service Info: Host: STEVE; OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for Seemi-s-A07 (192.168.10.64)
Host is up (0.00077s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE       VERSION
135/tcp open  msrpc         Microsoft Windows RPC
139/tcp open  netbios-ssn?
445/tcp open  microsoft-ds?
1 service unrecognized despite returning data. If you know the service/version, please submit the f
ollowing fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port139-TCP:V=7.991%I=7%D=10/21%Time=6AD8978F%P=x86_64-pc-linux-gnu%r(G
SF:etRequest,5,"\x83\0\0\x01\x8f");
MAC Address: 00:0C:29:B0:A0:2B (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.991%E=4%D=10/21%OT=135%CT=1%CU=31357%PV=Y%DS=1%DC=D%G=Y%M=000C2
OS:9%TM=6AD897A3%P=x86_64-pc-linux-gnu)SEQ(SP=101%GCD=1%ISR=10D%TI=I%CI=I%I
OS:I=I%SS=S%TS=U)SEQ(SP=104%GCD=1%ISR=10C%TI=I%CI=I%TS=U)SEQ(SP=106%GCD=1%I
OS:SR=109%TI=I%CI=I%TS=U)SEQ(SP=109%GCD=1%ISR=10A%TI=I%CI=I%TS=U)SEQ(SP=F8%
OS:GCD=1%ISR=FF%TI=I%CI=I%TS=U)OPS(O1=M5B4NW8NNS%O2=M5B4NW8NNS%O3=M5B4NW8%O
OS:4=M5B4NW8NNS%O5=M5B4NW8NNS%O6=M5B4NNS)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFF
OS:F%W5=FFFF%W6=FF70)ECN(R=Y%DF=Y%T=80%W=FFFF%O=M5B4NW8NNS%CC=N%Q=)T1(R=Y%D
OS:F=Y%T=80%S=O%A=S+%F=AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=B
OS:1C2A1A3%Q=)T3(R=Y%DF=Y%T=80%W=0%S=Z%A=O%F=AR%O=%RD=B1C2A1A3%Q=)T4(R=Y%DF
OS:=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=B1C2A1A3%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%
OS:F=AR%O=%RD=B1C2A1A3%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=B1C2A1A3%Q
OS:=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=B1C2A1A3%Q=)U1(R=Y%DF=N%T=80%
OS:IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=N)IE(R=Y%DFI=N%T=80%
OS:CD=Z)

Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for metasploitable3-ub1404 (192.168.10.91)
Host is up (0.00085s latency).
Not shown: 992 filtered tcp ports (no-response)
```

Figure 3 — Step 3: Service details of 192.168.10.64

What this screenshot shows: 192.168.10.64 runs only three services: 135/tcp Microsoft Windows RPC, 139/tcp netbios-ssn and 445/tcp microsoft-ds (the last two with fingerprints Nmap could not name exactly). Service Info reports OS: Windows. The machine is a lightly exposed Windows 10 host.

```
OS:I=I%SS=S%TS=U)SEQ(SP=104%GCD=1%ISR=10C%TI=I%CI=I%TS=U)SEQ(SP=106%GCD=1%I
OS:SR=109%TI=I%CI=I%TS=U)SEQ(SP=109%GCD=1%ISR=10A%TI=I%CI=I%TS=U)SEQ(SP=F8%
OS:GCD=1%ISR=FF%TI=I%CI=I%TS=U)OPS(O1=M5B4NW8NNS%O2=M5B4NW8NNS%O3=M5B4NW8%O
OS:4=M5B4NW8NNS%O5=M5B4NW8NNS%O6=M5B4NNS)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFF
OS:F%W5=FFFF%W6=FF70)ECN(R=Y%DF=Y%T=80%W=FFFF%O=M5B4NW8NNS%CC=N%Q=)T1(R=Y%D
OS:F=Y%T=80%S=O%A=S+%F=AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=B
OS:1C2A1A3%Q=)T3(R=Y%DF=Y%T=80%W=0%S=Z%A=O%F=AR%O=%RD=B1C2A1A3%Q=)T4(R=Y%DF
OS:=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=B1C2A1A3%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%
OS:F=AR%O=%RD=B1C2A1A3%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=B1C2A1A3%Q
OS:=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=B1C2A1A3%Q=)U1(R=Y%DF=N%T=80%
OS:IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=N)IE(R=Y%DFI=N%T=80%
OS:CD=Z)

Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for metasploitable3-ub1404 (192.168.10.91)
Host is up (0.00085s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT     STATE  SERVICE     VERSION
21/tcp   open   ftp         ProFTPD 1.3.5
22/tcp   open   ssh         OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp   open   http        Apache httpd 2.4.7 ((Ubuntu))
445/tcp  open   netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
631/tcp  open   ipp         CUPS 1.7
3000/tcp closed ppp
3306/tcp open   mysql       MySQL (unauthorized)
8181/tcp closed intermapper
MAC Address: 00:0C:29:08:C5:97 (VMware)
Aggressive OS guesses: Linux 3.2 - 4.14 (95%), Linux 3.8 - 3.16 (95%), Linux 3.10 - 4.11 (92%), Lin
ux 3.13 - 4.4 (92%), Linux 3.13 (91%), Linux 3.13 - 3.16 (91%), OpenWrt Chaos Calmer 15.05 (Linux 3
.18) or Designated Driver (Linux 4.1 or 4.4) (91%), Linux 4.10 (91%), Android 5.0 - 6.0.1 (Linux 3.
4) (91%), Android 8 - 9 (Linux 3.18 - 4.4) (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
 .
Nmap done: 3 IP addresses (3 hosts up) scanned in 33.78 seconds

┌──(root㉿V2310)-[/home/steve]
└─# 
```

Figure 4 — Step 3: Service details of 192.168.10.91

What this screenshot shows: 192.168.10.91 is a Linux machine (Metasploitable 3, Ubuntu 14.04) with notably old software: ProFTPD 1.3.5 on FTP, OpenSSH 6.6.1p1 (Ubuntu 2ubuntu2.13), Apache 2.4.7 on HTTP, Samba 3.X - 4.X on 445, CUPS 1.7 on IPP, and MySQL on 3306. Aggressive OS guessing rates Linux 3.2 - 4.14 at 95%, and Service Info reports Unix/Linux. The complete service scan of all three hosts took 33.78 seconds.

Step 4 — Vulnerability checking with NSE vuln scripts (on Kali).

Nmap's vuln script category tested every open service for known vulnerabilities. The full run took 1671.27 seconds (about 28 minutes), so the long output was searched for hits instead of being read line by line.

```
sudo nmap --script vuln 192.168.10.71 192.168.10.64 192.168.10.91 -oN p5-vuln.txt
```

```
┌──(root㉿V2310)-[/home/steve]
└─# sudo nmap --script vuln 192.168.10.71 192.168.10.64 192.168.10.91 -oN p5-vuln.txt
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-21 06:50 -0400
Nmap scan report for V2434 (192.168.10.71)
Host is up (0.00036s latency).
Not shown: 984 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
53/tcp   open  domain
80/tcp   open  http
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
88/tcp   open  kerberos-sec
111/tcp  open  rpcbind
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
2049/tcp open  nfs
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5985/tcp open  wsman
MAC Address: 00:0C:29:C4:FA:41 (VMware)

Host script results:
|_smb-vuln-ms10-054: false
```

Figure 5 — Step 4: Vulnerability scan of 192.168.10.71

What this screenshot shows: On 192.168.10.71 the web checks found nothing: the IIS server reported no stored XSS, no CSRF and no DOM-based XSS vulnerabilities, and the SMB check smb-vuln-ms10-054 returned false (not vulnerable). The 16 open ports are unchanged from the baseline.

```
┌──(root㉿V2310)-[/home/steve]
└─# grep -i "VULNERABLE" p5-vuln.txt
|   VULNERABLE:
|     State: VULNERABLE
|   VULNERABLE:
|     State: LIKELY VULNERABLE
|   VULNERABLE:
|   Service regsvc in Microsoft Windows systems vulnerable to denial of service
|     State: VULNERABLE
|       The service regsvc in Microsoft Windows 2000 systems is vulnerable to denial of service cau
sed by a null deference
|       pointer. This script will crash the service if it is vulnerable. This vulnerability was dis
covered by Ron Bowes

┌──(root㉿V2310)-[/home/steve]
└─#
```

Figure 6 — Step 4: Searching the full vuln-scan output for VULNERABLE hits

What this screenshot shows: The entire 28-minute scan output was searched with grep for the word VULNERABLE. Exactly one script reported hits: smb-vuln-regsvc-dos, which tests the Windows Remote Registry service for a denial-of-service flaw — it printed VULNERABLE

states, including one LIKELY VULNERABLE (unconfirmed) case. No other classic vulnerability was flagged anywhere in the output.

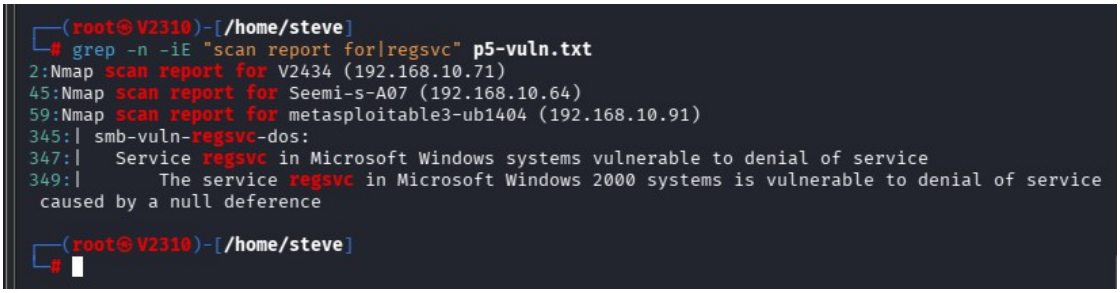


Figure 7 — Step 4: Locating which host the VULNERABLE hits belong to

What this screenshot shows: Line-number mapping proves the regsvc-dos hits sit inside the section for 192.168.10.91 (metasploitable3-ub1404, whose report begins at line 59; the script header is at line 345), i.e. against its Samba SMB service on port 445. The Windows hosts (.71 and .64, sections starting at lines 2 and 45) had no VULNERABLE hits at all. Against a Linux Samba target this check is likely a false positive, so the finding is recorded as low severity and unconfirmed — the crash test it describes was not executed.

4. Findings

#	Finding	Target	Severity	Evidence
F1	Remote Registry denial-of-service check flagged (smb-vuln-regsvc-dos reports VULNERABLE, one case LIKELY VULNERABLE)	192.168.10.91 (Samba on 445)	Low	Figures 6-7
F2	Outdated services with known historical CVEs: ProFTPD 1.3.5, OpenSSH 6.6.1p1, Apache 2.4.7, Samba 3.X-4.X, CUPS 1.7, MySQL	192.168.10.91	Medium	Figure 4
F3	Large attack surface: 16 open TCP ports	192.168.10.71	Low	Figures 1-2

	including FTP, DNS, LDAP, NFS and SMB on a domain controller			
F4	Minimal exposure: only 135/139/445 open; no vulnerability hits	192.168.10.64	Low	Figures 1, 3
F5	No web vulnerabilities found on IIS 10.0 (no XSS, no CSRF); smb-vuln-ms10-054 false	192.168.10.71	Info (hardened result)	Figure 5
F6	Firewall allow rule for port 8080 missing after the Project 4 cleanup — 8080 shows filtered instead of open	192.168.10.91	Low (config drift)	Figure 1

Severity notes: F1 is rated Low because it is a denial-of-service check on a legacy Windows service, reported against a Linux Samba target (likely a false positive), and one of the states is explicitly LIKELY VULNERABLE, i.e. unconfirmed. F3 is rated Low because a domain controller legitimately runs directory services; the risk is the breadth of the surface, not a confirmed hole — the earlier enumeration project proved anonymous access is denied (Project 3).

5. Remediation Roadmap

Priority	Action	Applies To
1	Patch or upgrade the outdated services on the Linux target (ProFTPD, OpenSSH, Apache, Samba, CUPS, MySQL) to currently supported versions	192.168.10.91
2	Restrict SMB (445) and other	All three targets

	management ports (22, 135, 139) to trusted hosts only, via host firewall rules	
3	Disable or firewall services that are not needed: FTP (21) and CUPS (631) if unused on .91; review NFS (2049) and FTP (21) on the domain controller	192.168.10.71, 192.168.10.91
4	Disable the Remote Registry service where it is not used, and block its access from untrusted networks	192.168.10.71, 192.168.10.64
5	Re-add the accidentally removed iptables allow rule for port 8080 on the Linux target (or confirm 8080 is meant to stay closed)	192.168.10.91
6	Keep the existing hardening: anonymous SMB enumeration restrictions on the domain controller, and unique per-service credentials everywhere	All three targets

6. Observations

- The assessment found no remotely exploitable critical vulnerability on any of the three targets: the Windows machines are patched and hardened, and the only script-flagged item is a likely false positive against Samba.
- The biggest real risk is the age of the software on 192.168.10.91 — several services are years out of support, and old versions are where future exploits land.
- The domain controller (.71) exposes 16 services, which is expected for its role, but every open service is a door that must be monitored and patched.
- The Windows 10 host (.64) is the quietest target: three open ports, no hits — the smallest attack surface of the three.
- Scanning discipline matters: the 28-minute vuln scan was searched with grep instead of read by hand, which found the single flagged script in seconds and proved the other two hosts were clean.

- The firewall change made during the Project 4 cleanup (three extra original rules removed by accident) was visible in this assessment as port 8080 changing from open to filtered — configuration drift is itself a finding (F6).

7. Result

Project 5 is complete. Three lab targets were assessed end to end: host discovery confirmed all three online; port scans mapped 16 open ports on 192.168.10.71, 3 on 192.168.10.64 and 6 on 192.168.10.91; service and OS detection identified every service and version; and the NSE vuln-script scan (about 28 minutes) found no critical vulnerability — the single flagged check (regsvc denial-of-service on the Linux target's Samba service) is recorded as a low-severity, likely false positive. The findings table and the six-step remediation roadmap above close the assessment. All five final projects are now finished.